Supplementary Material: Borrowing Strength Across Classes

Anonymous CVPR submission

1. Proofs

Throughout, $Q_\lambda = \alpha I + \lambda L_T$, $N_z = \mathrm{diag}(n)$, and for a fixed feature dimension j we write $d = d_{:j}$, $e = e_{:j}$ and $s = s_j$.

Derivation of ??. Given z, the terms of **??** that depend on D are $\sum_{i,k} z_{ik} \frac{(x_{ij} - t_{kj} - d_{kj})^2}{2\kappa\sigma_j^2}$ summed over j, plus $\frac{1}{2\kappa\bar{\sigma}^2} \sum_j d_{:j}^\top Q_\lambda d_{:j}$. Expanding the square around $\bar{x}_{kj}$ and dropping terms that do not depend on d gives, per dimension, $\frac{1}{2\kappa\sigma_j^2} \sum_k n_k (d_k - e_k)^2 + \frac{1}{2\kappa\bar{\sigma}^2} d^\top Q_\lambda d$. Multiplying by $\kappa\sigma_j^2$ and setting the gradient to zero yields $(N_z + s_j Q_\lambda) d = N_z e$, which is **??**. The matrix is positive definite whenever $\alpha > 0$ or $n_k > 0$ for some k in every connected component, so the solution is unique.

Proof of the special cases (??). With $\lambda = 0$ the system is diagonal: $(n_k + s\alpha) d_k = n_k e_k$. With $\alpha = 0$ and L_T the normalised Laplacian of the complete graph, $L_T = \frac{K}{K-1}(I - \frac{1}{K}\mathbf{1}\mathbf{1}^\top)$, whose null space is spanned by $\mathbf{1}$. As $\lambda \to \infty$ the minimiser of $\frac{1}{2}\sum_k n_k (d_k - e_k)^2 + \frac{\lambda s}{2} d^\top L_T d$ is forced into the null space, $d = \delta\mathbf{1}$, and minimising over δ gives $\delta = \sum_k n_k e_k / \sum_k n_k$. □

Proof of the harmonic-extension property (??). Partition the system by A and P. Rows in A have $n_k = 0$ and $e_k = 0$, so they read $s\,(Q_{\lambda,AA} d_A + Q_{\lambda,AP} d_P) = 0$, which gives the harmonic extension $d_A = -Q_{\lambda,AA}^{-1} Q_{\lambda,AP} d_P$ ($Q_{\lambda,AA}$ is positive definite when $\alpha > 0$, or when every component of A touches P). Substituting into the P rows gives $(N_{z,P} + s(Q_{\lambda,PP} - Q_{\lambda,PA} Q_{\lambda,AA}^{-1} Q_{\lambda,AP})) d_P = N_{z,P} e_P$, which is **??** on P with the Schur complement $Q_\lambda / Q_{\lambda,AA}$, the Kron reduction of the graph onto P [1]. □

Proof of the posterior-variance bound (??). Given z, the per-dimension posterior of d is Gaussian with precision $(N_z + sQ_\lambda)/(\kappa\sigma_j^2)$, so its covariance is $\kappa\sigma_j^2 (N_z + sQ_\lambda)^{-1}$. For symmetric positive definite A and positive semi-definite B, $(A + B)^{-1} \preceq A^{-1}$, hence the diagonal entries satisfy $[(N_z + s\alpha I + s\lambda L_T)^{-1}]_{kk} \le [(N_z + s\alpha I)^{-1}]_{kk} = 1/(n_k + s\alpha)$. Strictness: $[(A + B)^{-1}]_{kk} = [A^{-1}]_{kk}$ for all k in a set C would require B to vanish on the corresponding block of the Woodbury expansion, which fails for any class k with $(L_T)_{kk} > 0$, that is, any class with at least one neighbour. □

Proof of monotone block updates (??). With $\lambda_L = 0$ the objective is, in z, a sum of N strictly convex functions on the simplex whose minimisers are the softmax in **??** (standard mean-field argument), in π a Dirichlet-regularised multinomial likelihood whose minimiser is **??**, in D a strictly convex quadratic (above) and in σ^2 a separable problem whose stationary point is the stated variance. Each block update therefore does not increase J, which is bounded below, so J converges. With $\lambda_L > 0$ the term $-\frac{\lambda_L}{2m_s} z^\top W_s z$ is handled by the linear bound $-z^\top W_s z \le -2 z^\top W_s z^{(t)} + z^{(t)\top} W_s z^{(t)}$, valid when W_s is positive semi-definite and used as a majoriser otherwise, as in [3, 5]. □

2. Batch and online algorithms

Algorithm 1 GRAFT, batch (transductive) inference

Require: features $X \in \mathbb{R}^{N \times d}$, anchors $T \in \mathbb{R}^{K \times d}$, graph L_T, (α, λ) or empirical Bayes, c, κ, λ_y, λ_L

$\hat{y} \leftarrow \mathrm{softmax}(\tau X T^\top)$; $z \leftarrow \hat{y}$; $D \leftarrow 0$; $\pi \leftarrow \mathbf{1}/K$; $\sigma^2 \leftarrow$ variance of X about T under $\hat{y}$

$W_s \leftarrow m_s$-NN cosine graph of X

for $t = 1, \ldots, T_{\max}$ **do**

  E-step: update z by **??** (n_{inner} sweeps)

  $n \leftarrow z^\top \mathbf{1}$; $\bar{X} \leftarrow N_z^{-1} z^\top X$; $E \leftarrow \bar{X} - T$ (rows with $n_k = 0$ set to 0)

  $\pi \leftarrow (n + c)/(N + Kc)$ ▷ presence

  optionally $(\alpha, \lambda) \leftarrow \arg\max$ marginal likelihood of E (**??**)

  $D \leftarrow$ solution of **??** ▷ graph-pooled field

  $\sigma_j^2 \leftarrow \frac{1}{N}\left[q_j - 2\sum_k \mu_{kj} S_{kj} + \sum_k n_k \mu_{kj}^2\right]$ with $\mu = T + D$

end for

return z (posterior), (D, π, σ^2)

Algorithm 2 GRAFT, online inference on a stream

Require: anchors T, graph L_T, forgetting $\varrho \in (0, 1]$; state $(n, S, q, N) \leftarrow 0$, $D \leftarrow 0$, $\pi \leftarrow \mathbf{1}/K$

1: **for** each batch $X^{(b)}$ **do**
2: $\hat{y} \leftarrow \mathrm{softmax}(\tau X^{(b)} T^\top)$; E-step with current (D, π, σ^2) **??**
3: $(n, S, q, N) \leftarrow \varrho\,(n, S, q, N) + \big(z^\top \mathbf{1},\ z^\top X^{(b)},\ \sum_i x_i^{\odot 2},\ |X^{(b)}|\big)$
4: M-steps of Algorithm 1 from (n, S, q, N); re-run the E-step on $X^{(b)}$ (two rounds)
5: **return** posterior z of $X^{(b)}$
6: **end for**

3. Empirical Bayes details

Given the responsibilities, the empirical displacements of the present classes $P = \{k : n_k \geq 1\}$ are, under the model, $e_P = d_P + \epsilon$ with $\epsilon_k \sim \mathcal{N}(0, \kappa\sigma^2/n_k)$ and $d_P \sim \mathcal{N}(0, \tilde{Q}_P^{-1})$, where $\tilde{Q}_P$ is the Schur complement of Q onto P. With the isotropic approximation $\sigma_j^2 \approx \bar{\sigma}^2$, the marginal likelihood of E_P is a Gaussian with covariance $C = \kappa\bar{\sigma}^2(\tilde{Q}_P^{-1} + N_{z,P}^{-1})$ per dimension,

$$\log p(E_P \mid \alpha, \lambda) = -\tfrac{1}{2}\,\mathrm{tr}\,\big(E_P^\top C^{-1} E_P\big) - \tfrac{d}{2}\log\det C + \text{const}, \tag{1}$$

which is maximised over the grid $\alpha \in \{0.25, \ldots, 8\}$, $\lambda \in \{0, 1, \ldots, 32\}$. With P the present classes, the marginal covariance of $e_{:j,P}$ in pseudo-count units is $\tilde{Q}_P^{-1} + N_{z,P}^{-1}$ where $\tilde{Q}_P$ is the Schur complement of Q_λ onto P, because marginalising a GMRF onto a subset of nodes replaces the precision by its Schur complement [2]. The evidence of Eq. (1) sums the d independent dimensions after whitening E by $\sqrt{\kappa\bar{\sigma}^2}$. One Cholesky factorisation of a $|P| \times |P|$ matrix per grid point suffices; the Schur complement requires one solve with $Q_{\lambda,AA}$, which is sparse and well conditioned when $\alpha > 0$. The rule is disabled when $|P| < 3$, since the evidence is then nearly flat in λ.

4. Datasets and protocol details

Table 1. Benchmarks (CoOp test splits). K: vocabulary size; N_{test}: number of test images.

Dataset	K	N_{test}	Dataset	K	N_{test}
ImageNet	1000	50,000	Oxford Pets	37	3,669
SUN397	397	19,850	Flowers102	102	2,463
FGVC-Aircraft	100	3,333	Caltech101	100	2,465
EuroSAT	10	8,100	DTD	47	1,692
Stanford Cars	196	8,041	UCF101	101	3,783
Food101	101	30,300	ImageNet-V2 / S	1000	10,000 / 50,889
			ImageNet-A / R	200	7,500 / 30,000

The batch sampler draws K_{eff} uniformly in $[k_{\min}, k_{\max}]$, selects K_{eff} classes without replacement, pools their test images and draws B of them without replacement (all of them if fewer than B are available). The online sampler draws, for each class, Dirichlet(γ) proportions over K slots, assigns the class's shuffled images to slots accordingly, shuffles within slots and concatenates the slots; $\gamma = -1$ denotes the separate stream. These re-implementations follow the description of [4]; the released adapter also runs GRAFT inside the official harness so that the exact official samplers can be used.

5. Hyper-parameters

Symbol	Meaning	Value
τ	logit scale of the VLM	100
κ	likelihood temperature (inherited)	50
λ_y	exponent of the zero-shot prior (inherited)	1
λ_L, m_s	sample-graph weight and neighbours (inherited)	1, 3
α	anchor strength (pseudo-counts)	1
λ	graph coupling (pseudo-counts)	4
m, ρ	class-graph neighbours and bandwidth	5, 0.05
c	Dirichlet pseudo-count of π	0.5
$T_{\max}$, n_{inner}	outer and inner iterations	10, 3
n_{cg}	conjugate-gradient iterations	30
ϱ	online forgetting factor	1 (none)

6. Mechanism on synthetic data

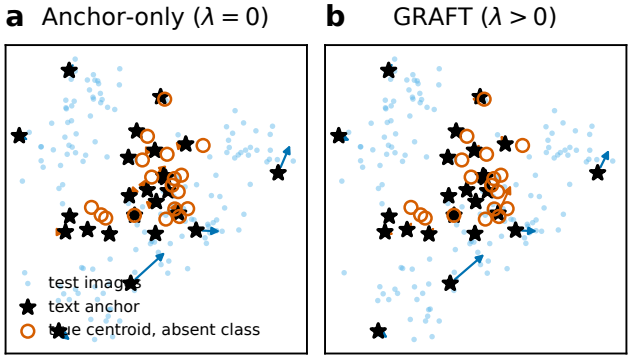


Figure 1. Adapted prototypes on synthetic data (6 of 24 classes present, PCA to two dimensions). Arrows go from text anchors to adapted prototypes; orange marks absent classes and circles their true centroids.

With $\lambda = 0$ absent classes stay at their anchors, whereas the pooled field moves them towards the centroids their images would occupy.

Table 2. Prior and graph ablation in the low setting (default: α=1, λ=4, m=5, ρ=0.05). Standard deviations over tasks are 0.3 for every entry.

α	0	0.25	1	4	16	
acc.	69.0	70.2	**70.7**	70.4	69.6	
λ	0	1	4	16	64	∞
acc.	69.1	70.1	**70.7**	70.5	69.9	68.8
graph	m=3	m=5	m=10	compl.	rand.	
acc.	70.4	**70.7**	70.5	68.8	69.2	

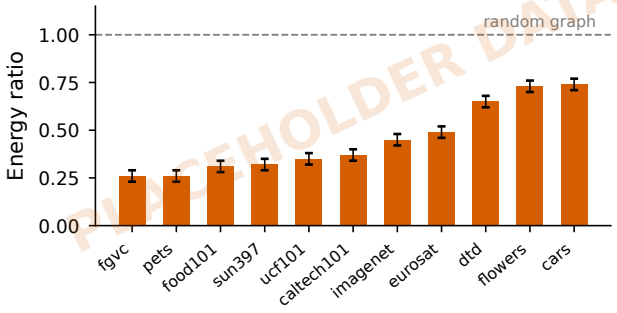


Figure 2. Class-level smoothness: Dirichlet energy of the oracle displacement field on the text graph relative to degree-matched random graphs (lower is smoother; error bars: standard deviation over 20 random graphs).

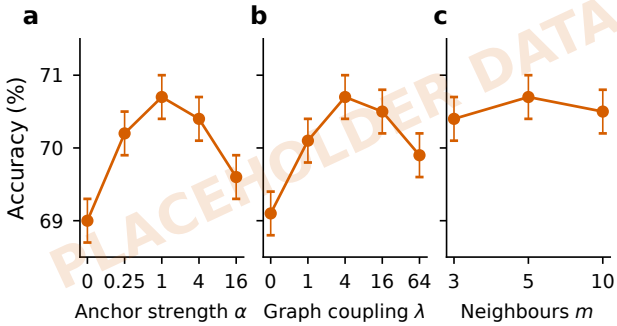


Figure 3. Sensitivity to the prior strengths and the graph size in the low setting (mean and standard deviation over tasks).

7. Prior and graph ablation

8. Smoothness and sensitivity plots

9. Additional result tables

10. Per-dataset results

Per-dataset versions of the main tables are generated by the released scripts (`visualization/generate_tables.py`) and will be included here once the runs complete.

Table 3. Backbones and distribution shifts, low setting (B=64, $K_{\text{eff}} \in [2, 10]$), top-1 accuracy (%). IN: ImageNet; V2/S/A/R: its shifted variants.

Backbone	Method	IN	V2	S	A	R
CLIP ViT-B/16	zero-shot	TBD	TBD	TBD	TBD	TBD
	StatA	TBD	TBD	TBD	TBD	TBD
	GRAFT	TBD	TBD	TBD	TBD	TBD
CLIP ViT-L/14	zero-shot	TBD	TBD	TBD	TBD	TBD
	StatA	TBD	TBD	TBD	TBD	TBD
	GRAFT	TBD	TBD	TBD	TBD	TBD
OpenCLIP B/16	zero-shot	TBD	TBD	TBD	TBD	TBD
	StatA	TBD	TBD	TBD	TBD	TBD
	GRAFT	TBD	TBD	TBD	TBD	TBD
SigLIP B/16	zero-shot	TBD	TBD	TBD	TBD	TBD
	StatA	TBD	TBD	TBD	TBD	TBD
	GRAFT	TBD	TBD	TBD	TBD	TBD

Table 4. Trustworthiness in the low setting (B=64), averaged over eleven datasets, CLIP ViT-B/16. ECE and NLL: lower is better; worst-decile accuracy: higher is better.

Method	ECE (%) ↓	NLL ↓	Worst-decile acc. (%) ↑
CLIP zero-shot	TBD	TBD	TBD
TransCLIP	TBD	TBD	TBD
StatA	TBD	TBD	TBD
MOON	TBD	TBD	TBD
Anchor-only control	TBD	TBD	TBD
GRAFT	TBD	TBD	TBD

Table 5. Likelihood, optimisation and cost, low setting. Time is the wall-clock per task of 64 images on one GPU with cached features (ImageNet, K=1000).

Variant	acc. (%)
default (shared diagonal Σ, no re-projection, c=0.5, κ=50, $T_{\max}$=10)	TBD
isotropic Σ	TBD
per-class diagonal Σ_k (as StatA)	TBD
re-project μ_k onto the sphere	TBD
$c \in \{0.1, 2\}$	TBD/ TBD
$\kappa \in \{25, 100\}$	TBD/ TBD
$T_{\max} \in \{3, 20\}$	TBD/ TBD
exact solver instead of conjugate gradients	TBD

Table 6. Computational cost: asymptotic cost per outer iteration and measured time per task (B=64, ImageNet, one GPU).

Method	dominant cost	ms / task
zero-shot	$O(NKd)$	TBD
TransCLIP	$O(NKd + Nm_sK)$	TBD
StatA	$O(NKd + Nm_sK)$	TBD
MOON	$O(NKd + Nm_sK)$	TBD
GRAFT (CG)	$O(NKd + Nm_sK + n_{\text{cg}}Kd(1+m))$	TBD
GRAFT + EB	above $+ O(\|G\|K^3)$	TBD

References

[1] Florian Dörfler and Francesco Bullo. Kron reduction of graphs with applications to electrical networks. *IEEE Trans-*

actions on Circuits and Systems I, 60(1):150–163, 2013. 1

[2] Håvard Rue and Leonhard Held. *Gaussian Markov Random Fields: Theory and Applications*. Chapman and Hall/CRC, 2005. 2

[3] Maxime Zanella, Benoît Gérin, and Ismail Ben Ayed. Boosting vision-language models with transduction. In *NeurIPS*, 2024. 1

[4] Maxime Zanella, Clément Fuchs, Christophe De Vleeschouwer, and Ismail Ben Ayed. Realistic test-time adaptation of vision-language models. In *CVPR*, 2025. 2

[5] Imtiaz Ziko, Jose Dolz, Eric Granger, and Ismail Ben Ayed. Laplacian regularized few-shot learning. In *ICML*, 2020. 1